

ATIVIDADE DE APOIO – ENSINO MÉDIO

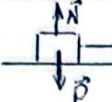
Componente Curricular: Física

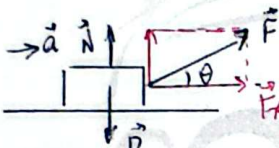
Professor (a): Rosana

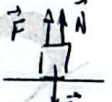
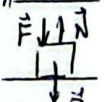
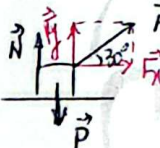
Assunto (s): Sistemas de Blocos - Ex - Modelos

Data: Setembro/25

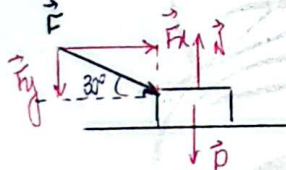
Série: 1ª

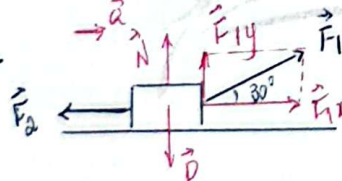
1.  a) Vertical $\Rightarrow$ repouso ($F_R = 0$)
 $N = P \Rightarrow |N| = 100 \text{ N}$ b) $F_R = m \cdot a \Rightarrow F = m \cdot a \Rightarrow 20 = 10 \cdot a$
 $a = 20/10 \Rightarrow |a| = 2 \text{ m/s}^2$

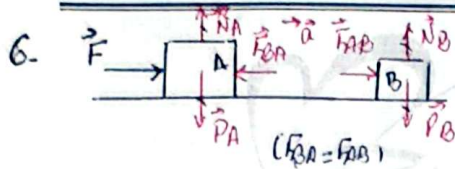
2.  a) Vertical $\Rightarrow F_R = 0$
 $N + F_y = P$
 $N + 60 = 200 \Rightarrow |N| = 140 \text{ N}$ b) $F_R = m \cdot a \Rightarrow F_x = m \cdot a$
 $80 = 20 \cdot a$
 $a = \frac{80}{20} \Rightarrow |a| = 4 \text{ m/s}^2$

3. a)  $N = P \Rightarrow |N| = 100 \text{ N}$
 $F + N = P$
 $30 + N = P$
 $|P| = 70 \text{ N}$
 $N = F + P$
 $N = 30 + 100$
 $|N| = 130 \text{ N}$
 $F_y = F \cdot \sin 30^\circ$
 $F_y = 30 \cdot \frac{1}{2} = 15 \text{ N}$
 $N + F_y = P$
 $N + 15 = 100 \Rightarrow |N| = 85 \text{ N}$

b) $F_R = m \cdot a \Rightarrow F \cdot \cos 30^\circ = m \cdot a \Rightarrow 30 \cdot \frac{\sqrt{3}}{2} = 10 \cdot a \Rightarrow a = \frac{3\sqrt{3}}{2} \text{ m/s}^2$

4.  a) $F_y = F \cdot \sin 30^\circ \Rightarrow F_y = 10 \cdot 0,5 \Rightarrow |F_y| = 5 \text{ N}$
 $N = F_y + P \Rightarrow N = 5 + 12 \Rightarrow |N| = 17 \text{ N}$
b) $F_x = F \cdot \cos 30^\circ \Rightarrow F_x = 10 \cdot 0,87 \Rightarrow |F_x| = 8,7 \text{ N}$
 $F_x = m \cdot a \Rightarrow 8,7 = 12 \cdot a$
 $a = \frac{8,7}{12} \Rightarrow |a| = 0,725 \text{ m/s}^2$

5.  a) $F_x = F \cdot \cos 30^\circ$
 $F_x = 20 \cdot \frac{\sqrt{3}}{2} \Rightarrow |F_x| = 17 \text{ N}$
 $F_R = m \cdot a$
 $F_x - F_2 = m \cdot a \Rightarrow 17 - F_2 = 50 \cdot 0,1$
 $F_2 = 17 - 5 \Rightarrow |F_2| = 12 \text{ N}$
b) $F_y = F \cdot \sin 30^\circ$
 $F_y = 20 \cdot \frac{1}{2} = 10 \text{ N}$
 $N + F_y = P$
 $N + 10 = 500 \Rightarrow N = 500 - 10$
 $|N| = 490 \text{ N}$

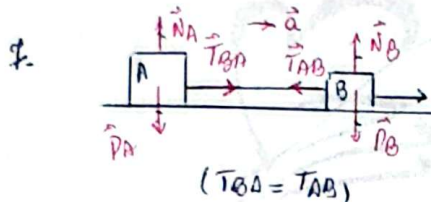


a) $\vec{F}_R = m \cdot a$
 $A \Rightarrow F - F_{BA} = m_A \cdot a$
 $B \Rightarrow F_{AB} = m_B \cdot a$
 $F = (m_A + m_B) \cdot a$

$10 = (6 + 4) \cdot a$
 $10 = 10 \cdot a \Rightarrow a = \frac{10}{10} \Rightarrow \boxed{a = 1 \text{ m/s}^2}$

b) $F_{AB} = m_B \cdot a$
 $F_{AB} = 4 \cdot 1 \Rightarrow \boxed{F_{AB} = 4 \text{ N}}$

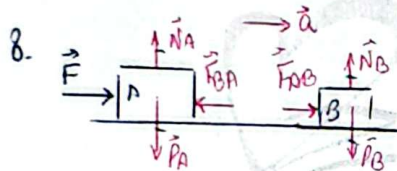
c) $\vec{F}_{RA} = m_A \cdot a$
 $F_{RA} = 6 \cdot 1 \Rightarrow \boxed{F_{RA} = 6 \text{ N}}$
 $\vec{F}_{RB} = F_{AB}$
 $\boxed{F_{RB} = 4 \text{ N}}$



$\vec{F}_R = m \cdot a \Rightarrow$
 $A \Rightarrow T_{BA} = m_A \cdot a$
 $B \Rightarrow F - T_{AB} = m_B \cdot a$
 $F = (m_A + m_B) \cdot a$

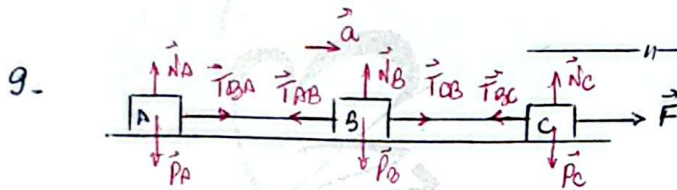
$4 = (5 + 3) \cdot a$
 $a = \frac{4}{8} \Rightarrow \boxed{a = 0,5 \text{ m/s}^2}$

$T_{BA} = m_B \cdot a \Rightarrow T_{BA} = 5 \cdot 0,5 \Rightarrow \boxed{T_{BA} = 2,5 \text{ N}}$



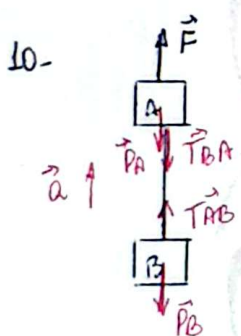
a) $\vec{F}_R = m \cdot a \Rightarrow$
 $A \Rightarrow F - F_{BA} = m_A \cdot a$
 $B \Rightarrow F_{AB} = m_B \cdot a$
 $F = (m_A + m_B) \cdot a$

$21 = (5 + 2) \cdot a$
 $a = 21/7$
 $\boxed{a = 3 \text{ m/s}^2}$
 b) $F_{AB} = m_B \cdot a$
 $F_{AB} = 2 \cdot 3$
 $\boxed{F_{AB} = 6 \text{ N}}$



$\vec{F}_R = m \cdot a \Rightarrow$
 $A \Rightarrow T_{BA} = m_A \cdot a$
 $B \Rightarrow T_{CB} - T_{AB} = m_B \cdot a$
 $C \Rightarrow F - T_{BC} = m_C \cdot a$
 $F = (m_A + m_B + m_C) \cdot a$

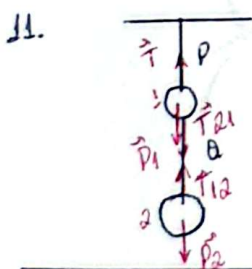
$6 = (2 + 5 + m_C) \cdot 0,6 \Rightarrow 10 = 7 + m_C$
 $\boxed{m_C = 3 \text{ kg}}$



$\vec{F}_R = m \cdot a \Rightarrow$
 $A \Rightarrow F - P_A - T_{BA} = m_A \cdot a$
 $B \Rightarrow T_{AB} - P_B = m_B \cdot a$
 $F - P_A - P_B = (m_A + m_B) \cdot a$

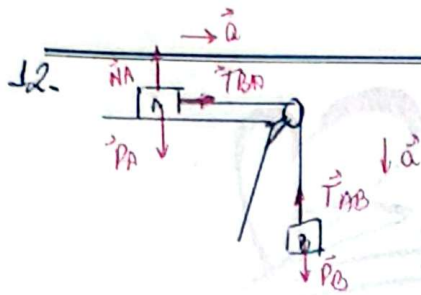
$F - 40 - 30 = (4 + 3) \cdot 2$
 $F - 70 = 14 \Rightarrow \boxed{F = 84 \text{ N}}$

$T_{BA} - 30 = 3 \cdot 2$
 $\boxed{T_{BA} = 36 \text{ N}}$



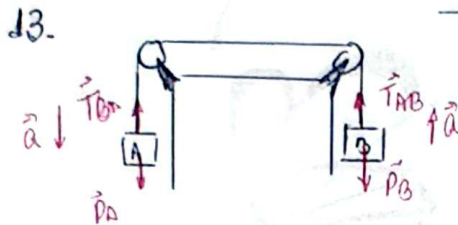
$\vec{F}_R = 0 \Rightarrow$ Repouso ($F_R = 0$)
 $1 \Rightarrow T = P_1 + T_{21}$
 $2 \Rightarrow T_{12} = P_2 \Rightarrow T_{12} = 20 \text{ N (for } a)$

$T = 10 + 20$
 $\boxed{T = 30 \text{ N (for } P)}$



$$\begin{aligned} A \rightarrow T_{BA} &= m_A \cdot a \\ B \rightarrow P_B - T_{AB} &= m_B \cdot a \\ P_B &= (m_A + m_B) \cdot a \end{aligned}$$

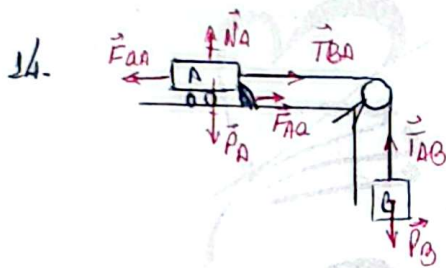
$$\begin{aligned} 10 &= (3+1) \cdot a & T_{BA} &= 3,25 \\ a &= \frac{10}{4} & |T_{BA} &= 15 \text{ N}| \\ |a &= 2,5 \text{ m/s}^2| \end{aligned}$$



* Dinamômetro indica 40N $\rightarrow T_{AB} = T_{BA} = 40\text{N}$
 * $P_A = 100\text{N}$ ($P_A > T_{BA} \therefore \downarrow a(A)$)

$$\begin{aligned} A \rightarrow P_A - T_{BA} &= m_A \cdot a \\ 100 - 40 &= 10 \cdot a \\ 60 &= 10 \cdot a \\ a &= 6 \text{ m/s}^2 \end{aligned}$$

$$\begin{aligned} B \rightarrow T_{AB} - P_B &= m_B \cdot a \\ 40 - 10 m_B &= 6 m_B \\ 16 m_B &= 40 \\ |m_B &= 2,5 \text{ kg}| \end{aligned}$$

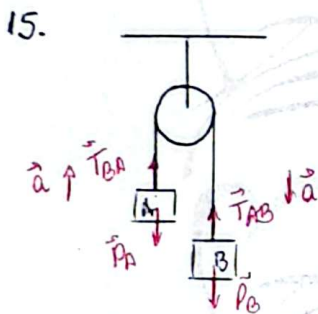


Repouso $\Rightarrow F_R = 0$

$$\begin{aligned} A \rightarrow T_{BA} &= F_{Aa} \\ B \rightarrow T_{AB} &= P_B \\ \therefore F_{Aa} &= P_B \Rightarrow |F_{Aa} = 50\text{N}| \end{aligned}$$

b) Sig idêntica à do Ex 12

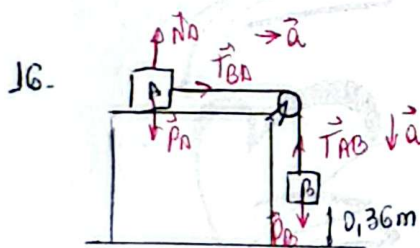
$$\begin{aligned} A \rightarrow T_{BA} &= m_A \cdot a \\ B \rightarrow P_B - T_{AB} &= m_B \cdot a \\ P_B &= (m_A + m_B) \cdot a \end{aligned} \quad \begin{aligned} 50 &= (20+5) \cdot a \\ a &= \frac{50}{25} = 2 \text{ m/s}^2 \end{aligned}$$



a) * $P_B > P_A$ $|B| \downarrow a$ $F_R = m \cdot a$

$$\begin{aligned} B \rightarrow P_B - T_{AB} &= m_B \cdot a \\ A \rightarrow T_{AB} - P_A &= m_A \cdot a \\ P_B - P_A &= (m_A + m_B) \cdot a \\ 120 - 80 &= (8+12) \cdot a \\ 40 &= 20 \cdot a \Rightarrow |a = 2 \text{ m/s}^2| \end{aligned}$$

$$\begin{aligned} b) P_B - T_{AB} &= m_B \cdot a \\ 120 - T_{AB} &= 12 \cdot 2 \\ T_{AB} &= 120 - 24 \\ |T_{AB} &= 96 \text{ N}| \end{aligned}$$



$$\begin{aligned} a) B \rightarrow P_B - T_{AB} &= m_B \cdot a \\ A \rightarrow T_{BA} &= m_A \cdot a \\ P_B &= (m_A + m_B) \cdot a \\ 400 &= (10+40) \cdot a \\ a &= \frac{400}{50} \Rightarrow |a = 8 \text{ m/s}^2| \end{aligned}$$

$$\begin{aligned} * T_{BA} &= 10 \cdot 8 \\ |T_{BA} &= 80 \text{ N}| \end{aligned}$$

$$\begin{aligned} b) \Delta S &= \frac{a}{2} t^2 \\ 0,36 &= \frac{8}{2} \cdot t^2 \\ t^2 &= 0,09 \\ t &= \sqrt{0,09} \\ |t &= 0,3 \text{ s}| \end{aligned}$$